Scientists Finally Recreate a 1,600-Year-Old Color-Changing Cup

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About 1,600 years ago, Roman glassmakers created a cup that changes colour like magic. Now, scientists have finally figured out how to make it again. The Lycurgus Cup, now on display at the British Museum, is one of the most famous pieces of Roman glass in the world. It is one of only 40 surviving glass **cage cups**. Look at it in reflected light and it appears jade green. Shine a light through it and it glows ruby red. Glassmakers have long wondered how the Romans pulled off this trick, called the **dichroic** effect. The leading explanation is that Roman craftspeople mixed tiny amounts of gold and silver into the glass. These metals form particles thousands of times smaller than a grain of sand, called **nanoparticles**. Because the particles are smaller than the **wavelengths** of light, they absorb and scatter certain colours depending on their size and shape. Lisa Hollands, a team leader at the University of Sheffield who has worked with glass for more than 25 years, set out to crack the recipe with Caroline Jackson, a leading expert on Roman glass. She tried different furnace types, starting materials with different oxygen levels, and even sugar. At last, she found a recipe that works: her glass looks opaque green in reflected light and glows red when light shines through. Hollands explained that the particles block blue and green light when light shines through, letting only reds and oranges pass. The team asked glass artist Elliot Walker, who won season two of Netflix's "Blown Away," to use the recipe in a new **sculpture**. The **sculpture** goes on display at the university's Festival of the Mind on November 6, before moving permanently into the school's Turner Museum of Glass. The science could go further than art: it may lead to new materials that control light, and to advanced glasses for safely storing **radioactive** waste.

Word Treasure

dichroic -- Dichroic means showing different colours in different light.
nanoparticles -- Nanoparticles are pieces of matter far too small to see.
cage cups -- A cage cup is a Roman glass cup with a delicate outer cage.
wavelengths -- A wavelength is the distance between two waves of light.
sculpture -- A sculpture is a work of art made by shaping material.
radioactive -- Radioactive means giving off energy from atoms breaking apart.

Background you need

The Romans were skilled glassmakers who made cups and vessels for wealthy people.

A museum is a place that keeps and shows important objects from history.

Nanoparticles are tiny pieces of matter, far too small to see, that can change how light behaves.

Scientists study old objects to learn how people in the past made things, and sometimes the same ideas lead to new materials.

Break it down

LEAD: Scientists finally recreate a 1,600-year-old colour-changing cup
KEY EVENT: Lisa Hollands cracks the Roman recipe
+- WHO: Hollands, Caroline Jackson and Elliot Walker
+- WHERE: University of Sheffield and the British Museum
+- HOW: Tiny gold and silver particles in the glass
EVIDENCE: The new glass looks green in reflected light and red when light shines through
OPEN QUESTION: What new materials could this science lead to?

Why it matters

Ancient crafts can teach modern science new tricks, and this one may help build materials that control light or store waste safely.

Test what you remember

1. What colours does the Lycurgus Cup show?

- (A) Red in reflected light and green when light shines through
- (B) Jade green in reflected light and ruby red when light shines through
- (C) White in reflected light and black when light shines through
- (D) Blue in reflected light and gold when light shines through

2. How old is the Lycurgus Cup?

- (A) About 600 years old
- (B) About 1,600 years old
- (C) About 16,000 years old
- (D) About 160 years old

3. What tiny metals in the glass cause the colour change?

- (A) Iron and copper
- (B) Gold and silver
- (C) Tin and lead
- (D) Zinc and nickel

4. Who cracked the recipe?

- (A) Christina Metcalfe at NASA
- (B) Lisa Hollands at the University of Sheffield
- (C) Elliot Walker at the British Museum
- (D) Caroline Jackson at the Vatican

5. What did Elliot Walker make with the new glass?

- (A) A sculpture
- (B) A mirror
- (C) A telescope
- (D) A window

6. When does the sculpture go on display?

- (A) At the Turner Museum on November 6
- (B) At the British Museum on November 6
- (C) At the Festival of the Mind on December 6
- (D) At the Festival of the Mind on November 6

Answer key (try the quiz first!): 1: B 2: B 3: B 4: B 5: A 6: D

Questions to think about

1. Lisa Hollands: says making this glass is complex and not just a case of adding nanoparticles to glass

MY THOUGHTS (at least 20 words):